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BIM-Based Indoor Navigation Using End-to-End Visual Localization and ARCore

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ABSTRACT

Traditional localization methods based on sensors such as Bluetooth and Wi-Fi struggle to achieve the required accuracy for fine-grained applications like AR navigation or robot localization. Accurate global position estimation and seamless AR navigation in dynamic indoor environments can significantly enhance the intelligence of indoor scene applications. This paper proposes an as-built BIM-based indoor navigation using End-to-End visual localization and ARCore. First, based on pre-captured RGB image sequences, a sparse point cloud of the indoor space can be constructed using Structure from Motion (SFM) methods, and the camera pose of each RGB image can be obtained. At the same time, image features are extracted in a multi-scale manner using end-to-end deep learning, and an image database is constructed for later scene retrieval and matching. Second, Building Information Modeling (BIM) model is constructed for a specific indoor scene, and an automated reconstruction of the indoor navigation topology map is performed based on this model. Finally, by giving a query image, the optimal matching image in the image database is obtained through scene retrieval methods. And the camera pose of the query image is iteratively optimized by multiple feature maps. Then path planning is performed based on the indoor navigation topology map with ARCore. In order to assess the effectiveness of the proposed method, Experimental are conducted for the evaluation of the global positioning estimation module and AR navigation module using both public datasets and our own collected data. The experimental results demonstrate that in publicly accessible datasets, the global positioning error consistently remains within the 20-cm range, with the majority of these datasets achieving a positioning error of 10 cm or less. Furthermore, the experimental results of AR navigation show that the method enables an approximate positioning accuracy of 10 centimeters in AR navigation within a length of 50 m.

1 | Introduction

Achieving high-precision positioning and navigation in indoor environments remains a significant challenge in the fields of computer vision, robotics, and indoor mapping. Traditional indoor positioning systems primarily rely on hardware sensors to estimate the position of mobile platforms. However, these methods typically provide positioning accuracy within the range of 1–5 m and require continuous device maintenance over long periods, which limits their effectiveness for applications requiring finer accuracy, such as AR navigation and robotics (Zafari, Gkelias, and Leung 2019). In contrast, visual-based positioning methods, leveraging the built-in cameras of smartphones and robotics devices, have emerged as a promising solution. These methods do

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not rely on the deployment of indoor sensor infrastructure and offer the potential for centimeter-level positioning accuracy. As a result, visual-based localization has become a key area of research in indoor navigation and computer vision (Castle, Klein, and Murray 2008; Middelberg et al. 2014). These approaches have the advantage of being highly maintainable and adaptable, which makes them particularly well-suited for applications in augmented reality (AR), where precise positioning is essential for seamless user experiences. Indoor navigation systems also require detailed topological maps of the environment to function effectively. Building Information Modeling (BIM) has been recognized as a valuable resource for generating indoor navigation maps due to its ability to represent both the physical and functional attributes of a building. BIM models contain detailed semantic information about indoor elements, such as floors, ceilings, walls, and furniture, which can be utilized to construct indoor topological maps (Tang et al. 2015; Kasprzak, Komninos, and Barrie 2013; Isikdag, Zlatanova, and Underwood 2013; Teo and Cho 2016). While BIM models provide a comprehensive representation of indoor spaces, challenges remain in integrating these models with real-time localization systems, particularly in environments where fine-grained geometry and semantic information are lacking.

To address these challenges, the Industry Foundation Classes (IFC) model provides a more detailed and semantically enriched representation of building elements. The IFC model includes explicit definitions of components such as storeys, doors, windows, and spaces, along with their spatial relationships (e.g., stairs connecting floors and walls containing openings). This information is critical for deriving accurate indoor topologies, which can be used for intelligent routing and AR navigation applications. By incorporating semantically enriched 3D models, it becomes possible to enhance navigation systems with more robust spatial awareness, enabling more accurate and contextually aware AR navigation experiences.

With the rapid advancement of end-to-end feature extraction methods, leveraging deep learning for image feature extraction and matching has provided a crucial foundation for achieving high-precision image matching in indoor scenes with repetitive textures and dynamic objects (DeTone, Malisiewicz, and Rabinovich 2018; Jing and Tian 2020). Simultaneously, in the process of urbanization, the construction of BIM models for newly built urban structures or the reverse reconstruction of BIM models for existing buildings is often necessary (Wang, Guo, and Kim 2019; Chen and Luo 2014). This process serves as a significant data source for generating indoor navigation maps. To address existing challenges, this study introduces an end-to-end indoor positioning and AR navigation framework anchored on as-built BIM models. This framework capitalizes on the inherent structure of BIM models to generate indoor navigation maps. Simultaneously, it integrates an end-to-end feature learning network to discern robust image features, thereby facilitating precise indoor positioning. Furthermore, by harnessing the computational prowess of mobile devices and incorporating ARCore, the framework ensures continuous indoor positioning and navigation. This methodology provides a cohesive solution for localization and navigation, streamlining everything from the generation of navigation maps to the localization and ongoing navigation processes. Such advancements have the potential to significantly enhance the efficacy and applicability of indoor localization and navigation strategies. This paper presents the following contributions:

- *Real-time indoor navigation framework*: This work introduces a computational framework for indoor localization and navigation, integrating BIM-based topological networks with scene feature computation, visual localization, and continuous navigation. It simplifies the process by autonomously handling all necessary tasks.
- *Autonomous navigation network generation from BIM*: An innovative method for creating navigation topological networks from BIM models is presented. It identifies walkable areas, establishes path nodes, and aligns the BIM model with multi-view scenes using a progressive registration approach, merging BIM and visual data for efficient localization.
- *Advanced pose estimation for dynamic scenes*: A novel encoder-decoder network is proposed for multi-scale feature extraction and camera pose estimation. Dynamic object detection is integrated to refine feature probability, and a weighted residual loss function ensures robust pose determination in dynamic indoor environments.

2 | Related Work

Indoor AR navigation is an emerging technology that offers a novel and intuitive way of navigating indoor environments. This section presents an overview of the current state-of-the-art methods in indoor AR navigation and highlights the challenges associated with indoor navigation.

2.1 | Indoor Localization

Infrastructure-based methods rely on a pre-installed infrastructure of beacons or Wi-Fi access points that emit signals to locate a device (Zafari, Gkelias, and Leung 2019). These methods include radio frequency-based techniques such as Wi-Fi (Bagosi and Baruch 2011; Chen et al. 2016), Bluetooth (Chawathe 2008; Zhuang et al. 2016), and ultra-wideband (UWB) (Prorok and Martinoli 2014). Wi-Fi-based indoor localization has been widely studied due to the ubiquitous nature of Wi-Fi access points. Several methods have been proposed to estimate the location of a mobile device using Wi-Fi signals, such as fingerprinting (Jang and Hong 2018), triangulation (Kim, Bong, and Kim 2011), and machine learning-based approaches (Song et al. 2019). Despite the high accuracy achieved by these methods, challenges such as signal interference and the need for frequent calibration remain. Bluetooth-based indoor localization has also gained attention due to the widespread use of Bluetooth Low Energy (BLE) beacons. BLE beacons are small, low-cost devices that broadcast their location information to nearby devices. Several methods have been proposed to estimate the user's location using BLE beacons, such as trilateration (De Blas and López-de 2017), fingerprinting (Luo and Hsiao 2019), and particle filtering (Malekzadeh et al. 2019). However, BLE-based methods for indoor localization still encounter challenges such as limited range and accuracy interference from other Bluetooth devices.

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UWB technology uses short pulses of radio waves to measure the time of flight between a mobile device and a UWB anchor. The distance between the device and anchor can be estimated by multiplying the time of flight by the speed of light. Several methods have also been proposed for UWB-based indoor localization (Prorok and Martinoli 2014; Heydariaan, Dabirian, and Gnawali 2020). Kim, Koo, and Kim (2023) evaluated the performance of mobile VIO data in enhancing absolute pose estimations in both desktop simulation and user study. However, still Indoor localization using those infrastructure-based methods still faces several challenges, such as multipath propagation problem, signal attenuation problem, cost and localization accuracy problem (Chen et al. 2023). More importantly, the localization accuracy of these device-based localization methods is maintained at around 1–5 m, which is difficult to achieve centimeter-level precision. However, the demand for location accuracy in existing indoor location services and robot applications is increasing, and relying solely on device-based localization methods is difficult to meet the demand for centimeter-level localization.

Visual-based methods do not rely on any pre-installed infrastructure and instead use sensors in a mobile device to estimate its position. Visual-based localization methods mainly rely on detecting and matching salient points or features in images (Ranganathan, Matsumoto, and Ilstrup 2013; Zhang, Liu, and Tian 2019), which potentially provides a centimeter-level localization accuracy. These methods extract features from images and then match them with the features from the reference images to compute the camera pose. Instead of calculating the features on images, some researchers aim to bypass the feature extraction and matching steps by directly minimizing the photometric error between the input images and a 3D model of the scene (Engel, Koltun, and Cremers 2017; Engel, Schöps, and Cremers 2014). These methods typically rely on dense image alignment techniques, such as optical flow or image warping, to estimate the camera pose. However, these methods are difficult in handling large inter-frame motion or scene depth variations, which may cause the optimization to fail. Additionally, they are more sensitive to image noise and are less robust to changes in lighting conditions. Acharya et al. (2019) proposed a model-based visual tracking approach for indoor localization using a 3D building model. This method matches line features in the image with those in the model, thereby achieving robust pose estimation which exhibits good robustness under conditions of motion blur and lighting variations. However, it is challenging for this method to re-estimate the position when location information is lost. Acharya et al. (2022) optimized the method with semantic information. To address the issues of changes in lighting, viewpoint, and occlusions during indoor localization, some researchers introduce the deep learning method to learn a mapping from images to poses directly (Ummenhofer et al. 2017; Sarlin et al. 2021). These methods directly calculate image features through deep learning and use them for pose estimation. Compared to hand-craft features, features obtained through deep learning may have better robustness and adaptability to dynamic scenes or light changes. Acharya et al. (Acharya et al. 2020) proposed a framework that uses synthetic images to estimate the camera pose of real images. A similar framework is optimized by Acharya et al. (Acharya, Khoshelham, and Winter 2019), which utilizes simulated images from 3D models for deep neural network training, followed by conducting pose regression experiments with real images. This method eliminates the need for SFM (SFM) model reconstruction, enabling real-time position estimation. However, this end-to-end pose recovery method can only achieve a localization accuracy of about 2 m. Higher localization accuracy necessitates models with a higher level of detail. Acharya, Tatli, and Khoshelham (2023) improved the method, proposing two data simulation strategies, S2R-PoseNet and R2S-PoseNet, to achieve end-to-end pose recovery. The experiments demonstrate that the best model is obtained through the real-to-synthetic style transfer, with a localization error of 0.31 m and an orientation error of 3.36°. However, the accuracy of this localization method may be influenced by the authenticity of the 3D models. Additionally, this method cannot be directly applied to areas that lack real texture coverage. Manglani (2023) focuses on an autonomous indoor navigation method for robots using a vision camera, employing the A* algorithm for pathfinding and continuous SLAM for navigation. It is important to note that it does not require path extraction or global positioning. Chen, Zhu, and Liu (2023) applied the PoseNet (Kendall, Grimes, and Cipolla 2015) method for depth estimation of image information. This approach introduced a pose rectified network, which estimates only the rotational transformation between two adjacent frames of images for the camera pose problem. By doing so, they improved the results of pose estimation, further optimizing these results with the loss from the pose rectified network. This method demonstrates the potential application of visual positioning in the field of depth estimation.

2.2 | Indoor Navigation

Map-based Navigation has been a subject of extensive research for numerous decades, encompassing a diverse array of algorithms and techniques devised to address various navigation-related challenges, including route planning, map matching, geocoding, and spatial indexing. Traditional algorithms, such as Dijkstra's algorithm (Dijkstra 1959) and Hidden Markov Model-based map matching (Lou et al. 2009; Newson and Krumm 2009; Atia et al. 2017), have been widely used in commercial applications due to their well-established theoretical foundations and efficient implementation. Contemporary advancements in machine learning and graph-based algorithms, such as graph neural networks (GNNs) and Graph Attention Networks (Veličković et al. 2018), have demonstrated considerable potential in enhancing the accuracy, efficiency, and scalability of map-based navigation systems. GNNs, influenced by the transformer architecture presented in "Attention is All You Need" (Vaswani et al. 2017), have been employed to tackle issues like route planning and map matching, whereas GATs have been utilized for node classification and edge prediction tasks. Furthermore, recent progress in natural language processing, exemplified by the pre-trained language model BERT (Devlin et al. 2019), has stimulated the creation of Graphormer (Ying et al. 2021), a transformer-based architecture designed to address problems involving graph-structured data. Graphormer extends the transformer architecture to function with graph-structured data, such as road networks in map-based navigation, and has demonstrated superior performance compared to traditional GNN architectures on multiple benchmark datasets,

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underscoring the promising potential of machine learning and graph-based algorithms in resolving intricate problems within the realm of map-based navigation. To expedite the acquisition of navigation networks, (Zhou et al. 2020) proposed an automated method for generating navigation networks based on BIM. This method involves extracting information like ceilings and walls from BIM models to construct occupancy grids, and then using a refinement algorithm to produce indoor topological maps. However, this method has only been validated for the automated generation of navigation paths based on BIM models and does not explore which elements in the automation process represent walkable spaces and which are obstacles. Additionally, this method has not been directly applied in real-time navigation frameworks.

AR-based navigation can benefit from different SLAM techniques, which enable robots to estimate their position and orientation while constructing a map of their surroundings (Huang, Zhao, and Liu 2020; Abaspur Kazerouni et al. 2022). Feature-based SLAM techniques represent the environment as a collection of unique landmarks or features detected and tracked by sensors like cameras or LIDAR. Algorithms such as EKF-SLAM (Smith, Self, and Cheeseman 1987), FastSLAM (Montemerlo et al. 2002), and ORB-SLAM (Mur-Artal, Montiel, and Tardos 2015) have gained popularity due to their computational efficiency and robustness to noise and occlusions, making them suitable for mobile robotics and low-power devices. ORB-SLAM3 (Campos et al. 2021), a state-of-the-art feature-based SLAM algorithm, further improves robustness and flexibility by incorporating dynamic object detection, global bundle adjustment, and multi-session mapping. Direct SLAM methods, on the other hand, use raw sensor data to estimate the device's position and map the environment without requiring feature detection and matching. Although an emerging research area, direct SLAM, exemplified by Direct Sparse Odometry (Engel, Koltun, and Cremers 2017), offers computational efficiency and proficiency in handling noisy and dynamic environments. However, questions remain regarding scalability and performance in large-scale environments. Geometric SLAM approaches employ geometry for pose estimation and environment mapping, incorporating techniques such as laser odometry, visual odometry, and bundle adjustment. While generally more accurate and robust than feature-based or direct SLAM, geometric SLAM can be computationally demanding and require extensive sensor data. State-of-the-art geometric SLAM algorithms, like Optimized Keypoint-based Visual-Inertial SLAM (Leutenegger et al. 2013) and Robust Visual Inertial Odometry (Bloesch et al. 2015), combine visual and inertial sensors to estimate the device's pose and construct dense 3D maps. These algorithms demonstrate resilience in challenging lighting conditions and motion dynamics, making them suitable for various environments, including large-scale outdoor scenes and indoor spaces.

Existing indoor localization methodologies predominantly concentrate on acquiring global positional data, often neglecting the intricate methodological and technical issues inherent in the combined processes of localization and navigation. Such issues encompass the automation of topological network construction and the direct utilization of these networks in indoor localization and navigation tasks. To address these gaps, this study introduces a comprehensive computational framework. This framework cohesively merges the creation of topological networks derived from BIM models, the computation of indoor scene attributes, feature map delineation, visual localization, and ongoing navigation. Beyond facilitating global positional computations, the framework is adeptly designed to bolster AR navigation applications, thereby simplifying the multifaceted nature of indoor localization and navigation.

3 | METHDOLOGY

3.1 | Overview

As shown in Figure 1, the overall framework of visual localization and AR navigation proposed in this paper is presented. It can be divided into three main parts. Firstly, based on pre-captured RGB image sequences, a sparse point cloud of the indoor space can be constructed using SFM methods, and the camera pose of each RGB image can be obtained. At the same time, image features are extracted in a multi-scale manner using end-to-end deep learning, and an image database is constructed

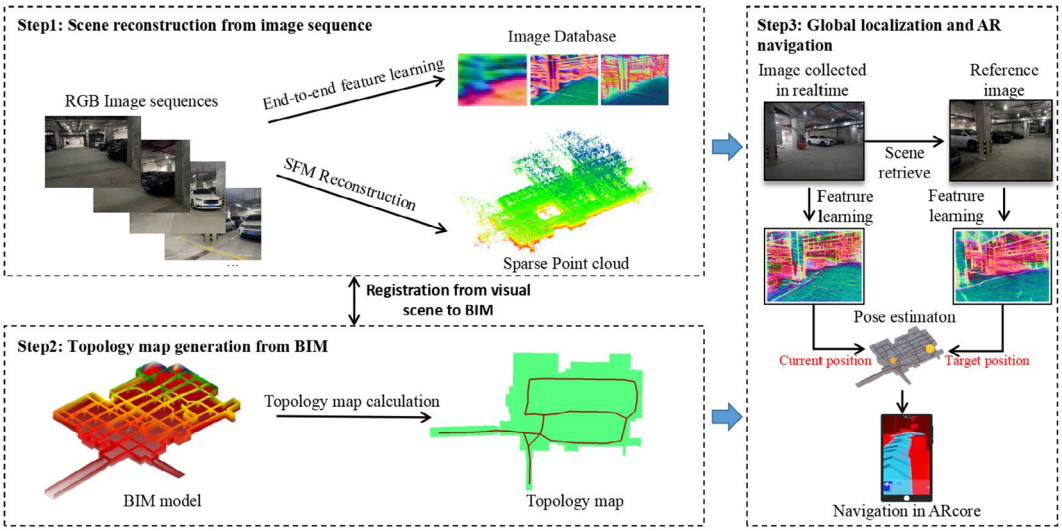

FIGURE 1 | Overview of the proposed framework.

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for later scene retrieval and matching. Secondly, a structured BIM model is constructed for a specific indoor scene, and an automated reconstruction of the indoor navigation topology map is performed based on this model, which will be used for later indoor path planning. Finally, in the global localization and real-time AR navigation part, the current location image is obtained through a mobile device, and the optimal matching image in the image database is obtained through scene retrieval methods. Multi-scale feature maps of the reference image and the query image are calculated using end-to-end feature computation methods, and the camera pose of the current location image is iteratively optimized. Based on this, users can input the target location, and path planning is performed based on the indoor navigation topology map, with the current location as the initial input, followed by real-time navigation using ARcore. The above methods achieve an integrated process of scene feature encoding, global position calculation, path planning, and real-time navigation.

3.2 | Scene Reconstruction From Image Sequence

3.2.1 | 3D Reconstruction and Poses Recovering

In this phase, we primarily employ the incremental SFM technique to acquire the camera poses and the sparse point cloud that signifies the underlying scene geometry. This algorithm fundamentally undertakes the task of reconstructing the three-dimensional structure by leveraging a sequence of images captured from diverse viewpoints. Within the incremental SFM framework, this process unfolds progressively, commencing from an initial image and gradually incorporating additional images, thus iteratively building up the comprehensive reconstruction. Therefore, in this step, the data input consists of a sequence of images captured in a specific scene, while the data output includes a sparse point cloud of the scene and precise pose information for each image. This process can be divided into three main components: feature extraction, feature matching, and geometric verification.

3.2.1.1 | Feature Extraction. In this paper, we utilize an improved COLMAP framework (Fisher et al. 2021) for sparse reconstruction of image sequences. Unlike the original COLMAP, we employ the SuperPoint feature detection algorithm for feature extraction in each image. In comparison to manually designed features, SuperPoint (DeTone, Malisiewicz, and Rabinovich 2018) has the ability to extract image feature information more effectively in texture-sparse regions or areas with repetitive textures. SuperPoint excels in descriptor-related metrics, such as nearest neighbor mean Average Precision (NN mAP) and matching score (M. Score), demonstrating that its learned feature representations for descriptor matching outperform those based on hand-crafted features (DeTone, Malisiewicz, and Rabinovich 2018). Moreover, it exhibits better robustness in the feature extraction and matching processes for large-scale scenes. Consequently, as shown in Equation (1), for each image, we obtain, F_i a sequence of local features, where x_j represents the local feature and f_j represents the key descriptor:

$$F_i = \left\{ \left(x_j, f_j \right) | j = 1 \ldots N \right\} \tag{1}$$

3.2.1.2 | Feature Matching. The primary objective of this phase is to address the challenge of feature point matching across images, ultimately yielding a collection of potentially overlapping images alongside their corresponding features. To accomplish this, we employ the SuperGlue algorithm, a feature matching approach based on graph convolutional neural networks (CNNs). SuperGlue is an extension of SuperPoint and leverages the feature points extracted by SuperPoint for matching purposes. In contrast to conventional matching methods based on feature descriptors, SuperGlue not only learns feature descriptors but also acquires a dedicated matching strategy that establishes correspondences between each feature point and all feature points in another image. Furthermore, it generates confidence scores for each matched pair. This learned matching strategy is automatically acquired through extensive training data, eliminating the need for manual design of a matching strategy.

3.2.1.3 | Geometric Verification. To address the issue of outliers in the matched pairs, where the matched feature points may not precisely correspond to points in the spatial domain. In this solution, RANSAC (Random Sample Consensus) is employed to eliminate mismatched points and enhance the accuracy of the essential matrix. Then the scene is reconstructed with an incremental reconstruction process. In the initialization step, a crucial decision is made regarding the selection of an initial pair of matching images, which significantly impacts the subsequent reconstruction. A higher number of matching points between the initial image pair leads to more robust and accurate reconstruction outcomes. Once the matching pair is identified, the camera poses of the two images are estimated using epipolar geometry, while simultaneously triangulating the corresponding 3D points. Moving forward, for each new image incorporated into the system, its camera pose is estimated through the Perspective-n-Point (PnP) algorithm, and additional 3D points are triangulated accordingly. To refine the reconstruction, Bundle Adjustment (BA) optimization is performed, effectively filtering out outliers throughout the iterative process, until all the images are seamlessly integrated into the system. This comprehensive approach ensures the overall accuracy and consistency of the reconstructed scene. The details of the incremental reconstruction are demonstrated in Algorithm 1.

3.2.2 | End-to-End Feature Learning for RGB Sequences

In this paper, an end-to-end visual feature computation method is employed to extract feature information from scene images. These features will be utilized to construct an image database and further utilized in the subsequent stages of image retrieval and matching processes. Drawing upon the seminal work presented in reference (Sarlin et al. 2021), our approach builds upon a ResNet-34 architecture, chosen as the backbone for feature extraction. This selection enables us to generate D-dimensional feature maps that effectively capture the essential information embedded within the input images. By incorporating diverse kernel sizes and multiple layers within the network, we achieve a comprehensive encoding of rich semantic details, facilitating the capture of intricate nuances while providing a holistic spatial context. Furthermore, a vital step in our approach involves the inclusion of an L2-normalization

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procedure applied to all extracted features before their output from the network. This critical normalization step not only enhances the robustness of our features but also significantly improves their generalization capabilities.

In the domain of long-term indoor scene localization, ensuring the utmost accuracy of the localization process can be hindered by changes occurring within the scene, such as furniture repositioning or object movements. Previous methodologies, exemplified by Sarlin et al. (Sarlin et al. 2021), have leveraged CNN (He et al. 2017) to assign confidence scores to features. However, this approach is susceptible to overfitting the training dataset, leading to suboptimal performance. To address this limitation, we propose a novel technique that incorporates Mask R-CNN for precise object segmentation within the scene. By accurately segmenting dynamic objects, we enhance the accuracy of feature point matching as well as the final localization outcome. Figure 2 provides a visual example showcasing the computation of confidence levels for a specific feature. P_i denotes the ith feature matching point within an image, w_i signifies the weight associated with this particular feature point, while w_q and w_{ref} encompass the weight maps of the query image and the reference image, respectively. The process commences with the system receiving an input image and conducting image retrieval to obtain a reference image. From the sparse point cloud data, we extract 3D scene points that have been registered within the reference image. These 3D points are then projected onto the feature maps of both the input and reference images, generating two corresponding vectors. The disparity between these vectors, known as the residual, serves as a quantification of the feature's confidence. This residual-based approach enables us to evaluate the reliability of individual features and improve the overall accuracy of the localization process:

ALGORITHM 1 | Incremental reconstruction.

Camera intrinsic parameters, Feature points, and Geometrically verified matching results.

3D point cloud, camera poses.
Calculate the tracks t of corresponding points.
Calculate the connected graph G (Nodes represent images, and edges represent that matching points).
Select $e \leftarrow G$.
Estimate the essential matrix E corresponding to e.
Decompose the essential matrix E to obtain the camera poses of the two images.
Triangulate the points in $t \cap e$ as the initial reconstruction result.
Delete e.
There are still e in G.
Extract e such that $track(e) \cap \{P_i\}$ is maximized.
PnP $\rightarrow$ estimate the camera pose.
Triangulate new *tracks*.
Delete e.
Bundle Adjustment.

$$w_i = \begin{cases} w_q{}^* w_r & \text{if } w_{\text{mask}} < 0.9 \\ 0 & \text{if } w_{\text{mask}} \geq 0.9 \end{cases} \tag{2}$$

To effectively handle dynamic objects in various scenarios, a pre-trained Mask-RCNN semantic segmentation network is employed. This network generates masks for dynamic objects, such as cars parked in a garage or people walking in a shopping mall. The specific computation for the weight w_i of a feature point is described in Equation (2). w_q denotes the probability prediction of the feature point after feature decoding, which is the product of the predicted probabilities of the corresponding feature point in both the reference image and the query image. w_{mask} represents the mask obtained through the dynamic object detection network. If the feature point lies within the mask, it indicates that the feature is part of a dynamic object, and the weight of the corresponding feature point is set to 0. By precisely segmenting dynamic objects using Mask R-CNN, their influence on feature point matching and localization can be effectively mitigated.

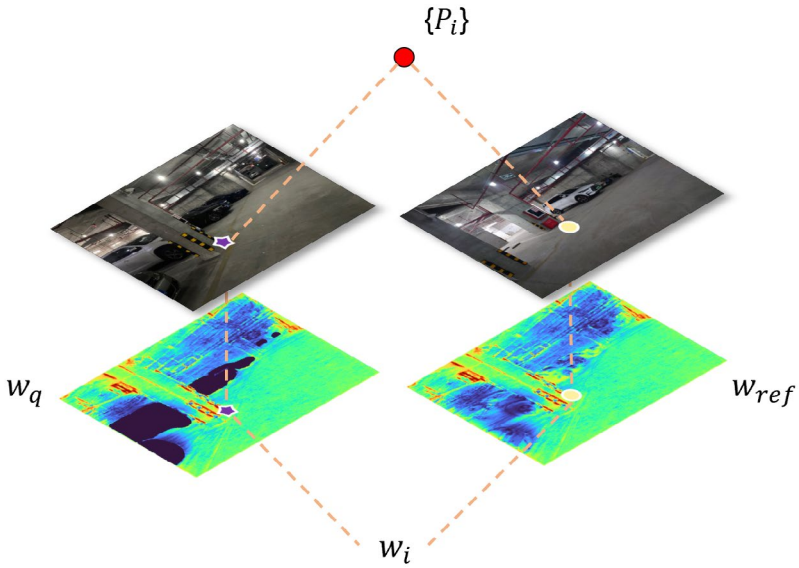

FIGURE 2 | An example illustrates how to estimate probabilities.

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3.3 | Topology Map Generation From BIM

BIM model is used for the generation of a topological map. The process of topology map generation from BIM is illustrated in Figure 3. The proposed framework initially performs semantic parsing of the BIM model, classifying and decomposing building elements such as floors, ceilings, beams, columns, and walls. To create a walkable area, all elements except for ceilings and beams are projected onto a 2D plane. The floor is identified as a potential walkable area, highlighted in green, while non-walkable regions like walls, columns, and parking areas are marked using a distinct color. By overlaying the walkable area with the floor using Boolean operations, a walkable area map is obtained.

Next, the central axis skeleton nodes are determined within the walkable area map utilizing a spatial fast refinement algorithm (Zhang and Suen 1984). To ensure accurate navigation, optimal topological nodes are selected based on spatial distance constraints. Finally, a topological graph, known as the human-path graph, is constructed using these nodes. This graph is then mapped back to the 3D map, enabling the generation of a comprehensive topological map for navigation purposes. The key steps involved in this process include generating a walkable area map and employing a fast refinement algorithm to determine the central axis skeleton nodes. The following will provide a detailed description of the walkable map generation and space skeleton generation methods.

3.3.1 | Walkable Map Generation

To facilitate navigation within the internal pathways of a building, it is essential to construct a topological map that accurately represents the real walkable areas. The proposed approach involves the following steps:

- *Classification and projection*: The components of the BIM model are classified based on semantic labels. These components are then projected onto a 2D plane, resulting in a classified image depicting the indoor walkable area. Semantic labels categorize the indoor elements into three types: (a) Walkable elements: ground-level components that form the foundation of the walkable areas. (b) Obstacle elements: elements like walls, columns, vehicles, and parking areas that obstruct movement within indoor spaces. These elements are spatially connected to the ground plane. (c) Unrelated elements: elements such as beams, small clutter, doors, and windows that have no direct impact on the extraction of interior access space.
- *Generation of walkable area map*: Walkable elements are projected onto the 2D plane, generating an initial binary image where green pixels represent the walkable area. Obstacle elements are then projected onto the same plane, and their corresponding pixels are filled with yellow. A map of the walkable area is created by applying Boolean operations. In this map, road pixels are filled with white.
- *Refinement algorithm*: The road pixels and black background of the walkable area map are merged, resulting in a binary image. This refined binary image serves as the input for the subsequent refinement algorithm.

3.3.2 | Skeleton Generation

The refinement process aims to extract the central axis or skeleton of an image by iteratively removing its outermost pixels, thereby reducing the stored information by transforming multi-pixel lines into single-pixel skeletons. The algorithm employed is an 8-neighbor parallel refinement algorithm. The steps involved in this process are as follows:

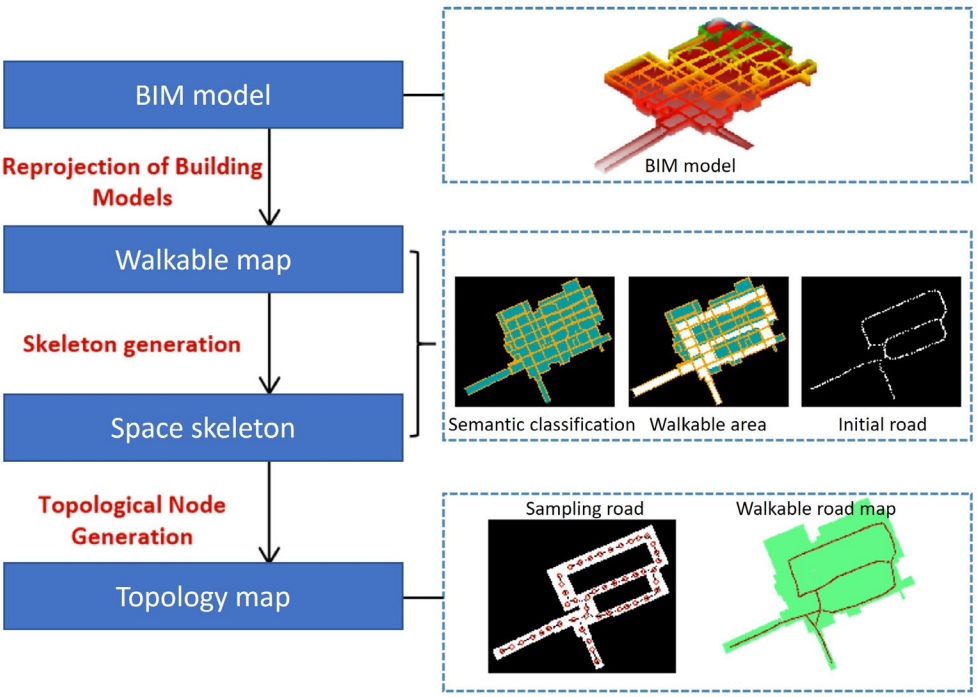

FIGURE 3 | Topology map generation from BIM.

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- *Boundary POINT DETECTION*: Initially, the algorithm identifies boundary points in the image. A boundary point is defined as a pixel labeled as 1 that has at least one neighboring pixel labeled as 0. Figure 4b illustrates the boundary points, and those satisfying Condition 1 are marked.
- Condition 2 Evaluation: The marked boundary points are retained and not deleted until the second operation step, which involves evaluating condition 2, as depicted in Figure 4c.
- *Iterative deletion*: The marked points undergo an iterative process where they are deleted if condition 2 is satisfied. This iterative algorithm continues until no further points can be deleted.

By following this refinement algorithm, the resulting output is a set of skeleton points representing the central axis of the image. These skeleton points can be utilized to determine optimal topological nodes based on spatial distance constraints. Ultimately, this process enables the generation of topological maps for navigation purposes.

3.4 | Global Localization

3.4.1 | Registration From BIM to SFM Scene

As shown in Figure 5, to align the scanned BIM model with the coordinate system generated by SFM, a combination of feature-based registration and the iterative closest points (ICP) algorithm (Besl and McKay 1992) is utilized.

- *Feature selection*: In the 3D scene point cloud, points with distinctive features like corners, edges, doorways, etc., are manually marked for selection. These feature points are roughly matched with their closest points in the BIM model.
- *Rotation matrix calculation*: After obtaining three pairs of corresponding points in both models, the approximate rotation matrix from the BIM model to the virtual 3D scene point cloud can be calculated. This rotation matrix is then applied to the entire BIM model.
- *ICP alignment*: The fine adjustment and alignment of the two models are performed using the ICP algorithm. The algorithm takes two point sets, namely the reference set S_{ref} and the new set S_{new}, as inputs. Additionally, the maximum number of iterations $(\max_i\text{terations})$ and a convergence tolerance value (tolerance) are defined. The initial transformation consists of an identity rotation matrix (R) and a zero translation vector (t), while the initial error is set to infinity and the iteration counter is set to zero.
- Iterative Optimization: In each iteration, correspondences between points in S_{new} and their closest points in S_{ref} are computed. Then, the optimal rotation matrix (R) and translation vector (t) are estimated to minimize the distance between corresponding points. The transformation is applied to update S_{new}, and the error between the reference and transformed new point sets is calculated. The iteration counter is incremented.
- *Convergence check*: The iterative process continues until the error falls below the specified tolerance or the maximum number of iterations is reached.
- *Output*: Upon completion, the algorithm returns the transformed point set S_{new} along with the computed transformation parameters R and t.

By following these optimized steps, the feature-based registration and ICP alignment method effectively align the BIM model with the virtual 3D scene, enabling accurate fusion and visualization of the two models.

3.4.2 | Image Retrivel

Our global localization method in the framework employs an iterative optimization process that progresses from known pose images to unknown pose images. To begin with, the first step is to perform image retrieval and identify keyframes in the given

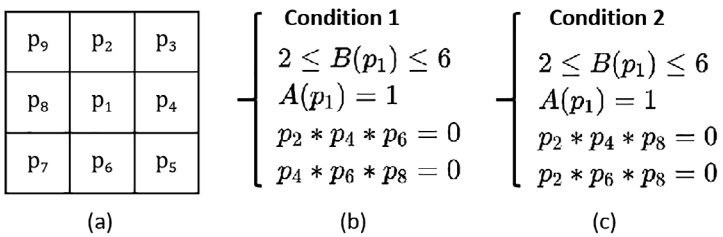

FIGURE 4 | Skeleton generation from walkable map. (a) 3 × 3 sliding window. (b) Demonstration of Condition 1. (c) Demonstration of Condition 2.

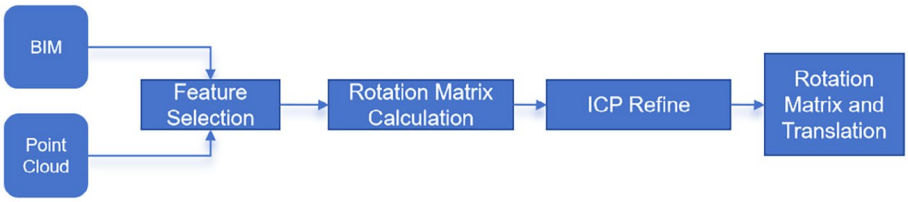

FIGURE 5 | Registration process.

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image database that belong to the same scene as the given image. This is achieved by extracting scene features from the images and converting them into feature vectors. Subsequently, the similarity of these feature vectors in a high-dimensional space is used as a measure of image similarity. The classical approach for image retrieval consists of the following steps: Given an image database I, extract the feature $f(I_i)$ for each image I_i using a feature function. Provide a query image q and extract its feature $f(q)$ using the same feature function. Compute the similarity between the query image feature $f(q)$ and the image database features $f(I)$ using a similarity measure, typically the Euclidean distance $D(q, I) = \|f(q) - f(I)\|$. A smaller distance indicates greater similarity between the images. To perform global feature extraction from images, we utilize the NetVlad method for image feature encoding. In this approach, we aggregate feature maps from the convolutional layers of a CNN into a fixed-size vector that represents the entire image. The calculation process is depicted in Equation (3):

$$V(j,k) = \sum_{i=1}^{N} \frac{e^{w_k^T x_i + b_k}}{\sum_{k'} e^{w_{k'}^T x_i + b_{k'}}} \left(x_i(j) - c_k(j)\right) \tag{3}$$

In the equation, $V(j, k)$ represents the element at position (j, k) in the aggregated vector. The sum is taken over N feature maps, where x_i denotes the feature map at index i. w_k and b_k are the learnable parameters associated with cluster k, while $c_k(j)$ represents the jth element of the cluster center for cluster k. The numerator computes the importance weight for each feature map, and the denominator normalizes the weights. Finally, the difference between each feature map element $x_i(j)$ and its corresponding cluster center $c_k(j)$ is multiplied by the weight and summed up to generate the final aggregated feature vector.

3.4.3 | Camera Localization With Deep Learning Features

After performing image retrieval, we proceed to calculate feature maps and confidence maps at three different scales for these images. The camera pose recovery is then accomplished by optimizing Equation (4):

$$L(R, t) = \sum \omega_k^i \rho\left(\left|r_k^i\right|^2\right) \tag{4}$$

In this equation, ω_k^i denotes the per-residual weight for the ith reference image. These weights are derived from the Uncertainty Map, a byproduct of the CNN training process. During training, the CNN assigns confidence scores to each pixel based on feature relevance and consistency in the prediction process. Pixels with higher confidence scores have a greater influence on the final pose estimation. The symbol ρ represents a robust cost function designed to reduce the network's sensitivity to outliers and noise. To achieve this, a soft inlier count method is employed, which filters out potential erroneous matches and improves the overall resilience of the camera localization. Lastly, the variable r_k^i represents a residual function that quantifies the discrepancy between the projections of the query image in 3D space and those of the reference image.

3.5 | Route Planning and AR Navigation

In this section, our focus lies on the integration of the route planning module with the AR navigation method, which serves as a pivotal aspect of our system. By seamlessly merging the route planning module with AR navigation, our objective is to deliver a cohesive and intuitive navigation experience that capitalizes on the synergistic potential of both the virtual and physical domains.

3.5.1 | Route Planning

To strike a balance between efficiency and effectiveness, we have opted to implement the Dijkstra algorithm as our primary route-planning approach within the system.

The simplified Dijkstra's algorithm starts by initializing a priority queue denoted as Q to store vertices, and two arrays, dist and prev, to hold distance and predecessor values for all vertices in the graph, respectively. The distance of the source vertex is set to 0, while all other vertices are assigned an initial distance value of infinity. Subsequently, all vertices are added to the priority queue with their respective distance values as keys.

The main loop of the algorithm continues until the priority queue becomes empty. In each iteration, the vertex u with the minimum distance value is extracted from the queue. If the distance of u is found to be infinity, the loop is terminated as no further reachable vertices remain to be processed. For each neighboring vertex v of the extracted vertex u, the algorithm computes an alternative distance, denoted as alt, by adding the distance of u to the weight of the edge connecting u and v. If this alternative distance is smaller than the current distance value of v, the distance and predecessor arrays are updated with the new values, and the key of v in the priority queue is also updated accordingly. The algorithm continues to iterate until all vertices have been processed or the priority queue becomes empty. Finally, the algorithm returns the distance and predecessor arrays, which can be utilized to determine the shortest path between the source vertex and any other vertex in the graph.

3.5.2 | Indoor Navigation Based ARCore

ARCore, developed by Google, is a platform designed to create AR experiences by leveraging the capabilities of the user's mobile device, including motion capture, environment understanding, and light estimation (Google 2024). To ensure accurate mapping of objects in the virtual 3D space, ARCore utilizes a clustering method to identify flat surfaces or textureless areas from the extracted feature points. This allows for determining the overall pose of the 3D space and subsequently calculating the object's position within the virtual environment. ARCore also incorporates depth information captured by the device to better control the size and scale of the virtual object, enhancing the realism of the visual experience.

The software system consists of two main components: the front-end and the back-end. The front-end includes a user

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interface, a sensor listener module, a 3D rendering engine, and a SLAM module. The back-end comprises an end-to-end positioning module and a path planning module, as depicted in Figure 6.

Initiated by the user through the interface, the sensor listener module captures image data from the device and utilizes an Application Programming Interface (API) to format the information. This data is then transmitted to the end-to-end positioning module in the back-end. The end-to-end positioning module extracts features from the image, matches them with pre-constructed scene models, and returns pose information associated with the image. Simultaneously, the path planning module receives destination information and calculates the shortest path to the designated destination. The path planning module returns a sequence of the shortest paths leading to the destination. The front-end independently processes the received pose and path information. The pose information is used to initialize the SLAM model, while the shortest path information is matched with paths in the topology graph, identifying the qualified paths. As the device moves, the SLAM module continuously updates the user's location in real-time on the interface.

4 | Experimental Analysis

4.1 | Data Collection and Preprocessing

4.1.1 | Datasets

To verify the effectiveness and accuracy of the localization and navigation framework for indoor space, we utilized two sets of data: the 7scenes public dataset (add reference) and our own collected dataset. The 7scenes dataset contains image sequences of indoor spaces from seven different areas, with image resolutions of 640 × 480 pixels and varying numbers of images ranging from 2000 to 14,000. It consists of indoor and outdoor environments in seven different scenes, such as office rooms and kitchens, with rich texture and structural features. Each scene includes RGB image sequences from multiple viewpoints and corresponding camera pose information. The self-collected dataset from an underground parking lot was specifically collected for this study to simulate complex indoor environments found in real-world scenarios. Our self-collected dataset was captured using a Google Pixel 4XL for reference image acquisition and an iPhone 11 for query image collection in parking lots with vehicle changes. The underground parking lot dataset presents challenging aspects, including low lighting conditions, limited texture, and complex structures. This allows for the evaluation of visual pose estimation and AR navigation systems in realistic and complex indoor environments. The specific information of the data is shown in the table below, and example images from different datasets are shown in Figure 7.

4.1.1.1 | Data Preprocessing. Each set of experimental data needs to undergo preprocessing before the accuracy evaluation can be conducted. As shown in Figure 8, given a sequence of images from a scene, we divide them into reference image sets and query image sets. The former is used to establish a basic description of the scene, while the latter is used for evaluating the positioning accuracy. To evaluate the accuracy of global localization, each set of data undergoes two

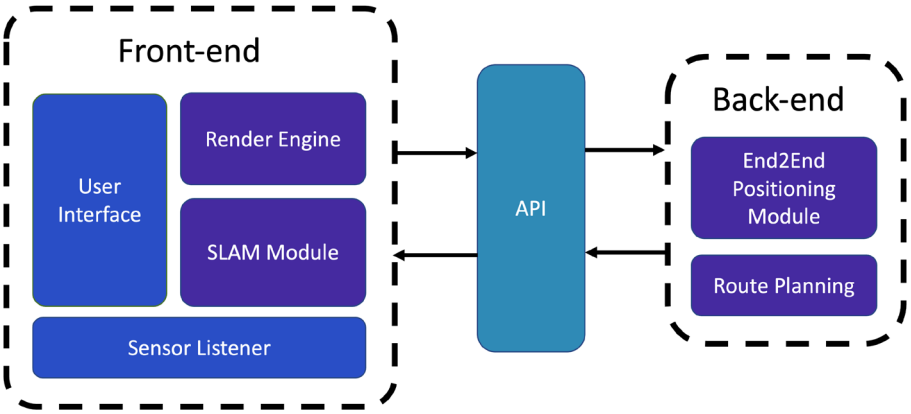

FIGURE 6 | Basic framework of the AR navigation in our solution.

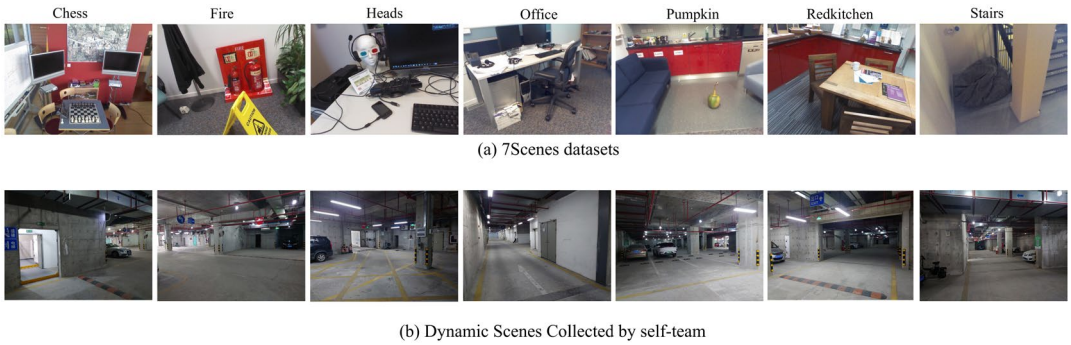

FIGURE 7 | Datasets, where (a) refers to publicly available datasets and (b) refers to self-collected datasets.

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structure-from-motion (SfM) processes. In the first process, all reference images are used for matching and sparse point cloud reconstruction. In the second process, the poses of the reference images are fixed, and the query images are included for SfM computation to recover the camera poses of the query images. After these steps, the pose information of the query images can be used as ground truth for accuracy evaluation.

4.1.1.2 | Experimental Setup. In order to conduct effective experimental validation, the 7scene dataset was used to assess the accuracy and robustness of the end-to-end visual pose estimation method proposed in this paper. Self-collected datasets were used to evaluate the visual localization of large-scale indoor scenes and the precision of AR navigation. It is worth mentioning that, during the experiments of AR navigation, we collected trajectory data for different walking distances and further verified the robustness of AR navigation under different lighting conditions by adjusting the lighting. Figure 9 illustrates the experimental setup utilized in this study. Initially, the feature extraction network is trained using the ImageNet dataset, facilitating the establishment of a global position estimation model. During the experimental phase, the study employs the 7Scenes dataset along with a self-collected dataset to validate the accuracy of global localization and navigation. Specifically, Section 4.2 evaluates the global localization accuracy of the two datasets separately, while Section 4.3 verifies the precision of AR navigation.

4.1.1.3 | Training and Implementation Details. Training and evaluation of the end-to-end localization method is conducted with a platform that consisted of eight NVIDIA A100 Tensor Core GPUs, each with 40GB memory. The PyTorch library was utilized to implement the basic framework of the network. We employed a pre-trained VGG16 model that was trained on the ImageNet dataset to extract CNN features and also incorporated a pre-trained Swin Transformer model due to the limitation of available training data. In this study, initial values were determined using pre-trained weights derived from the COCO dataset, followed by fine-tuning procedures implemented on the Extended CMU Seasons dataset (Sattler et al. 2018). This dataset encompasses a total of 7159 reference images and 75,335 query images, distributed across 17 distinct sections, illustrating substantial seasonal variations. Notably, these variations often highlight natural structures such as trees, thereby posing a notable challenge for effective feature matching. The analytical approach involved extracting co-visible image pairs to simulate the relative localization of one image to another, leveraging observed 3D points as a basis. This optimization procedure was iteratively performed for 15 cycles at each level, with the initial parameters set using the pose data from the reference image. Training on the dataset was structured methodically, utilizing sections 8–12 and 22–25 for the primary training phase, and sections 6, 13, and 21 earmarked for validation purposes. The training employed batches consisting of three image pairs, with each image resized to maintain a minimum dimension of 720 pixels. Moreover, square crops measuring 720 pixels were sampled, establishing the query pose inline with the reference image pose, promoting analytical consistency and precision. Regarding AR navigation, we developed an Android app with the following basic workflow: the app uploads images of the current location from the smartphone to a server. The server then automatically utilizes a visual localization algorithm to perform initial position calculation. Subsequently, continuous localization and navigation are carried out using ARCore. Therefore, for the 7scene dataset, we solely assessed the accuracy and effectiveness of its visual localization. For the underground parking dataset, we will evaluate both global localization and the precision of AR navigation simultaneously.

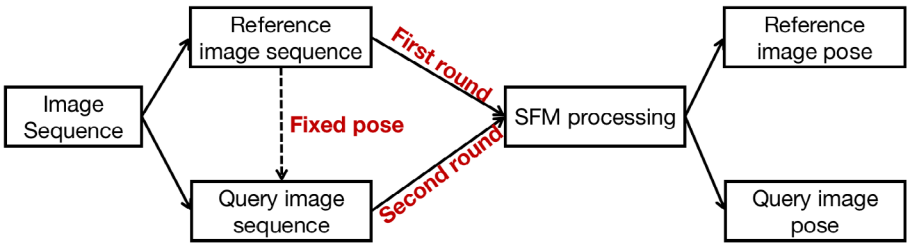

FIGURE 8 | Data preprocessing.

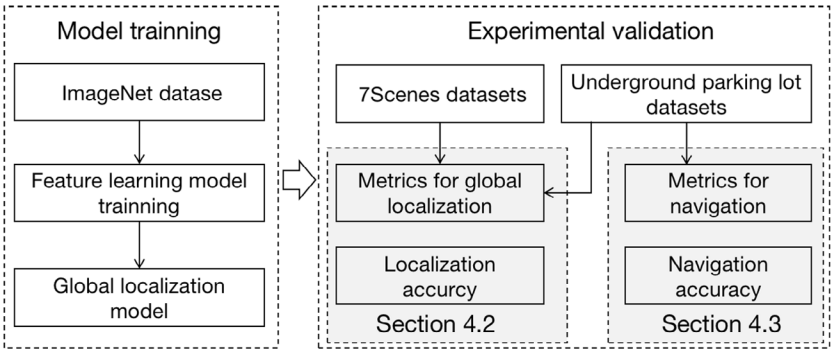

FIGURE 9 | Experimental setup.

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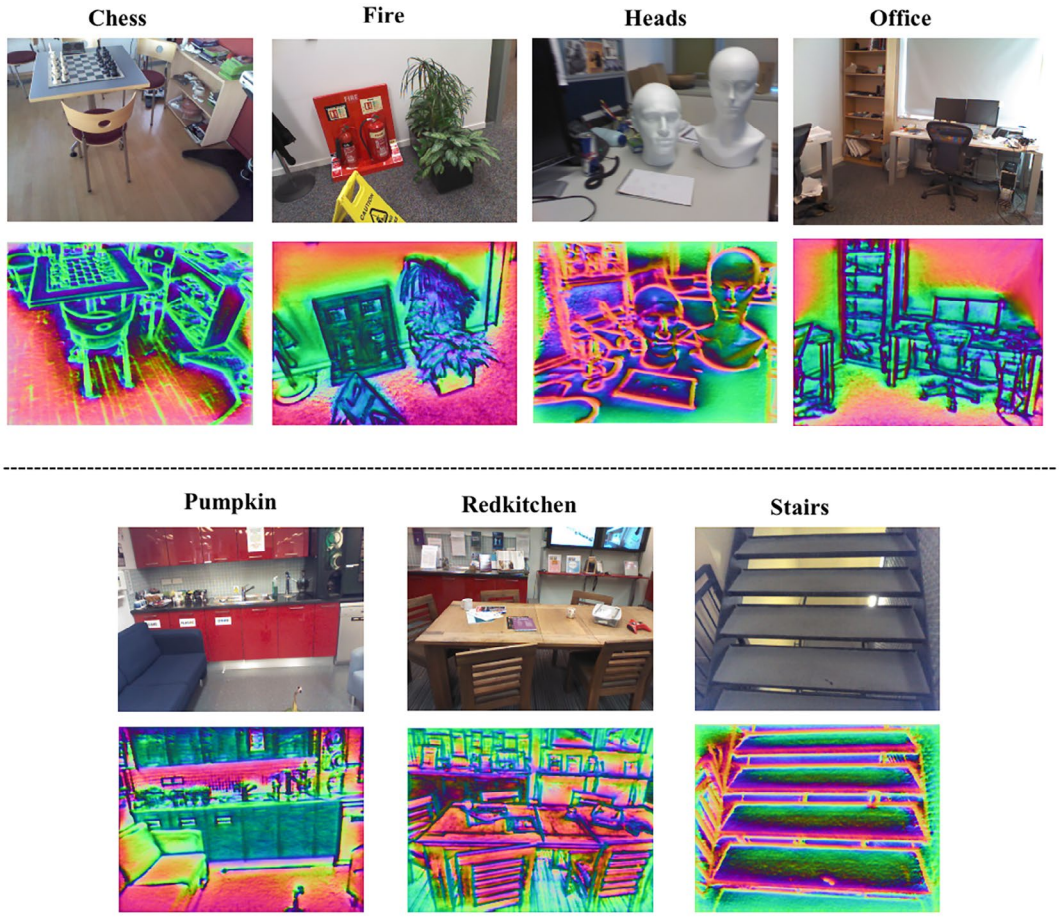

FIGURE 10 | Feature maps of the selected image in 7Scenes datasets.

4.2 | Accuracy Assessment of Global Localization

4.2.1 | Metrics for Accuracy Evaluation

In this study, to evaluate the accuracy of the global pose estimation method, this study calculates both the angular $\varepsilon_{\text{angle}}$ and positional errors $\varepsilon_{\text{position}}$ obtained from the algorithm. The angular error $\varepsilon_{\text{angle}}$ is computed by determining the cosine value between the estimated rotation matrix R and the ground-truth rotation matrix R_{gt}. Subsequently, it is further refined using the inverse trigonometric function. It is pivotal to note that in the event of an unsuccessful prediction, the angle error is assigned a value of 180°. Conversely, the positional error $\varepsilon_{\text{position}}$ is derived by gauging the Euclidean distance between the estimated coordinates t and the ground-truth coordinates t_{gt}. Therefore, The computation of $\varepsilon_{\text{angle}}$ and $\varepsilon_{\text{position}}$ are illustrated in Equations (5) and (6), respectively:

$$\varepsilon_{\text{angle}} = \arccos\left(\frac{\text{trace}\left(RR_{gt}^{T}\right) - 1}{2}\right) \tag{5}$$

$$\varepsilon_{\text{position}} = \| t - t_{gt} \| \tag{6}$$

4.2.2 | Results of 7Scenes Datasets

In order to validate the effectiveness of the global localization method proposed in this paper, we conducted accuracy evaluations on seven sets of data from the 7scenes dataset. Figure 10 showcases the selected image from each of the seven datasets, and the corresponding feature maps generated. The proposed end-to-end image feature calculation method effectively preserves the linear features of the images. Moreover, within the planar regions, the algorithm demonstrates consistent feature representations in local areas. The preservation of linear features and the consistency of planar features contribute to enhanced robustness and accuracy in feature matching between images. The aligning process between reference images and query images in the 7scenes datasets is illustrated in Figure 11. It is evident that the iterative optimization process aligns feature points in the scene effectively, resulting in accurate pose estimation for query images. Figure 12 presents our accuracy in localizing both position and orientation. It is depicted as a cumulative histogram of errors across the entire testing dataset. The proposed method achieves a localization accuracy of over 0.4 m in more than 80% of the query images across all scenes, highlighting its significant advantage in high-precision position estimation. Additionally, the angular errors remain within 10°

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FIGURE 11 | Alignment results of 7scenes datasets.

in all scenes, with the chess dataset showing remarkable performance, where 80% of the image angular errors are within 5°. Median errors indicate that all scenes achieve positioning accuracy at the 10-cm level. Notably, the chess and office datasets achieve position accuracies of 5 and 7 cm, respectively. Similarly, the median angular errors for all scenes are within 10°, with chess and office datasets exhibiting even better results. Based on the aforementioned analysis, the proposed global localization method achieves a positioning accuracy at the level of 10 cm. The minimal position and angular errors make it highly suitable for applications with stringent requirements for localization accuracy, such as AR.

In this study, the introduced method is scrutinized alongside two other leading visual localization frameworks, demonstrating marked superiority when applied to public datasets. Firstly, PoseNet (Kendall, Grimes, and Cipolla 2015), the pioneering method in this area, delegates the comprehensive procedure from feature extraction to final pose estimation to a singular trained model. This strategy, however, manifests a constrained generalization capability, necessitating retraining of the network when adapting to a different dataset. Contrastingly, the current work innovatively reconstructs a series of feature extraction architectures, diverging markedly from the approaches outlined in the recently delineated PixLoc (Sarlin et al. 2021). This redevelopment engenders enhanced stability in edge information and fortifies the robustness of feature extraction results. It can be observed that the PoseNet method, a type of end-to-end pose restoration technique, is markedly inferior to pose recovery algorithms based on feature points, exhibiting the weakest performance among the three methods assessed. It can be asserted that the pose estimation based on feature points significantly surpasses end-to-end approaches such as PoseNet in terms of localization accuracy. In comparison with the PixelLoc method, the approach proposed in this paper still maintains a pronounced advantage. Especially in the aspect of high-precision localization performance, approximately 80% of the images in the chess, heads, and office datasets can achieve a localization accuracy of 0.2 m using the method proposed in this paper, whereas PixelLoc can only achieve this level of accuracy in about 20% of the images. The method proposed in this manuscript significantly outperforms the other two methods in terms of angular accuracy, primarily due to two key innovations. First, the dynamic object exclusion mechanism optimizes feature weighting by filtering out the influence of moving objects in the environment. This exclusion prevents dynamic elements—such as pedestrians, vehicles, or machinery—from introducing noise into the feature extraction process, allowing the model to focus on stable, reliable features that are essential for accurate pose estimation. Second, the use of progressive localization refines pose estimation incrementally, improving angular accuracy. Instead of calculating the final pose in a single step, the process is broken down into smaller, iterative refinements based on incoming data. This allows the system to continuously adjust and correct its pose predictions, leading to more precise and stable results, especially in complex or uncertain environments. These two approaches—dynamic object exclusion and progressive localization—collectively enhance the accuracy and robustness of the proposed method.

4.2.3 | Results of Underground Parking Lot

To validate the effectiveness of our proposed visual pose estimation method in large-scale indoor scenes, we conducted accuracy verification using a self-collected dataset from an underground parking lot. This particular scene has distinct characteristics: it spans an area of 50 m by 30 m, exhibits a high degree of texture similarity across different areas, and includes dynamic objects. The presence of these dynamic and changing scenes, combined with the high degree of texture similarity, poses significant challenges for visual pose estimation. In Figure 13, we present an example image from the underground parking lot and demonstrate the generation of feature maps at three levels, along with feature weight maps. Similar to the 7scenes dataset, the feature maps in the parking lot maintain strong linear features. Moreover, they exhibit good feature consistency on both the floor and pillar planes. By analyzing the feature weight maps, we can observe that our algorithm effectively handles dynamic objects and applies mask processing to reduce their influence on the weight maps.

As shown in Figure 14, we analyzed a series of images taken from various perspectives within the underground parking lot, revealing the considerable variation in vehicle distribution and positions. Notably, we applied a mask processing technique to dynamically handle objects, resulting in lower weights assigned to the pixels corresponding to vehicles in the weight maps. Remarkably, even in scenarios where the car shapes are incomplete or occupy a small portion of the image, our approach achieves impressive results. Figure 15 illustrates the cumulative histogram of localization errors in the underground parking lot

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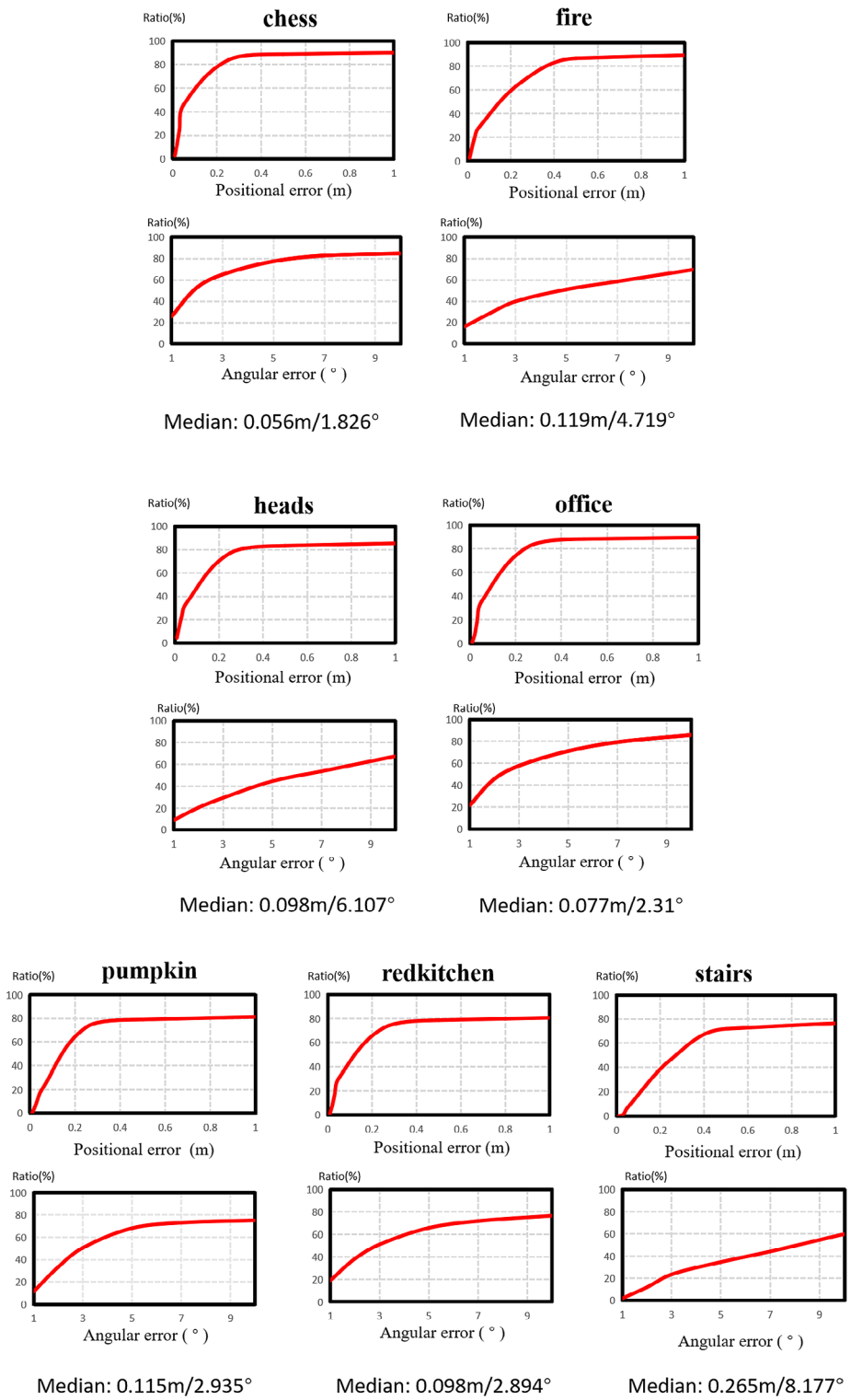

FIGURE 12 | Localization performance of 7scenes datasets.

environment. It can be observed that in the large-scale parking lot environment, the proposed localization algorithm achieves a median positioning error of approximately 3.5 cm. Specifically, around 76.5% of the images achieve a positioning accuracy within 5 cm, approximately 80% of the images achieve a positioning accuracy within 10 cm, and 90% of the images achieve a positioning accuracy within 20 cm. This demonstrates that the proposed global visual localization algorithm can achieve high positioning accuracy in large-scale dynamic scenes with sparse texture features. It possesses significant application potential for AR navigation in such scenarios.

4.3 | Performance of AR Navigation

In order to evaluate the accuracy and robustness of the AR navigation framework proposed in this paper, we collected four distinct sets of data from the parking lot dataset. These sets include

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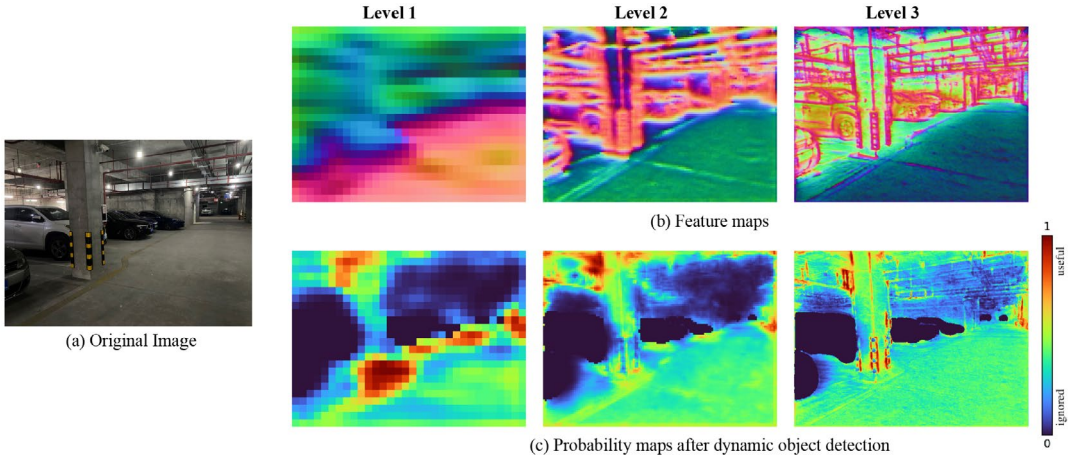

FIGURE 13 | Feature maps and weight maps in different scales.

FIGURE 14 | Weight maps of different views.

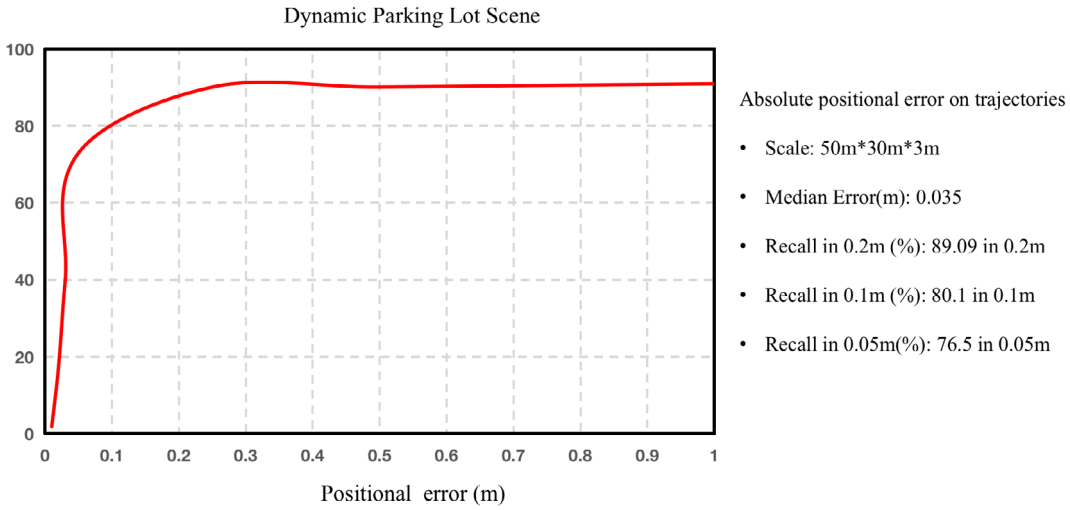

FIGURE 15 | Localization performance of dynamic parking lot scene.

loop closures with single-trip diameters of 5, 10, and 20 m, as well as a long-distance walking loop closure. For the experiments with distances of 5, 10, and 20 m, we collected data five times for each distance. As for the long-distance experiment, we collected data for two sets.

The data collection procedure for validating the AR navigation algorithm is illustrated in Figure 16. In each experiment, a standardized apparatus is utilized, strategically placed at diverse locations. The setup comprises a stationary tripod anchored to the ground and testing equipment, secured with a reliable latch

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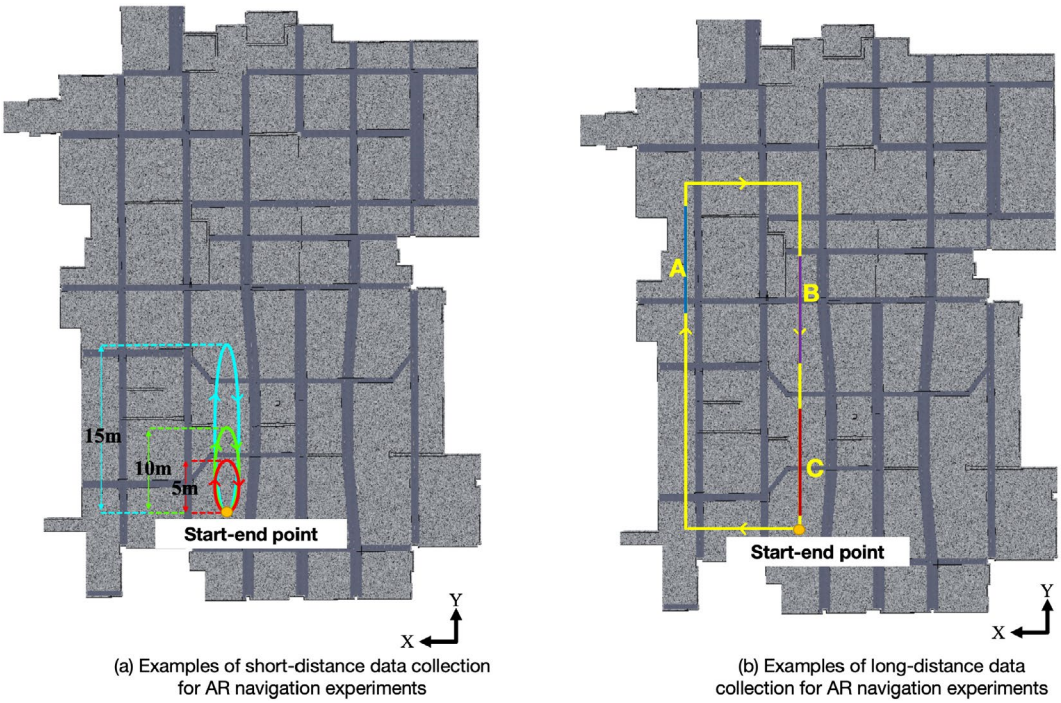

FIGURE 16 | Data collection for AR navigation evaluation.

Distance	5m	10m	20m	50m
Total walking distance	10m	20m	40m	150m
	Positional errors (m)			
1	0.0424	0.0224	0.0245	0.5917
2	0.0173	0.0245	0.0316	1.3391 (Add perturbations)
3	0.0244	0.0332	0.0361	NAN
4	0.03	0.0245	0.03	NAN
5	0.0141	0.05	0.1122	NAN

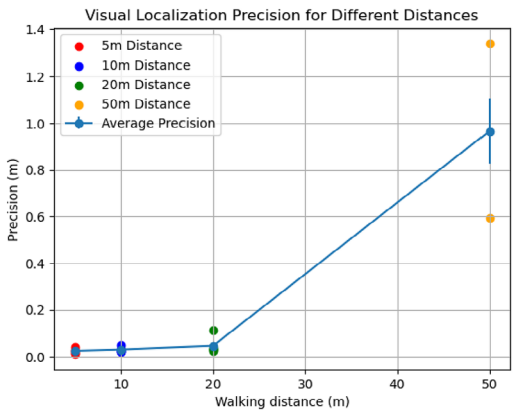

FIGURE 17 | The performance of AR navigation module in dynamic parking scenarios of different walking ranges.

mechanism, ensuring the consistency of the coordinate reference system by aligning the endpoint coordinates with the initial ones after each cycle. At the initiation of each trial, the testing equipment is unlatched, permitting a tester to manually handle it to accurately replicate real-life walking scenarios. Before the conclusion of each test, the equipment is re-secured to the tripod, marking the end of the trial when the readings stabilize, signaling a return to the original position. Throughout this process, the sophisticated positioning software diligently captures all the coordinate points delineated by the testing equipment's trajectory. Utilizing this dataset, positional discrepancies are quantified by computing the Euclidean distances between the recorded initial and final coordinates, thereby enabling a precise calculation of positional errors. It is worth noting that Figure 15b displays three regions labeled as A, B, and C. In the second round of data collection of the long-distance walking dataset, during the collection in region A, there were dynamic vehicle targets passing through the scene. In region B, the mobile device experienced local and intense shaking. In region C, we manipulated the lighting conditions by turning the lights on and off to introduce changes in the scene illumination. Based on these experiments, the robustness and adaptability of the AR navigation system can be verified across different environmental scenarios.

Figure 17 illustrates the AR navigation results of four experimental groups. The first three groups conducted five data collections each, while the last group performed two data collections. The distances traveled in these experiments were approximately 10, 20, 40, and 150 m, respectively. From the table in the figure, it can be observed that the AR navigation errors of the first three groups remained within 10 cm, and the errors were relatively stable across each experiment. In the first experiment of the long-distance fourth group, the loop-closure error of the

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AR navigation reached around 60 cm. However, in the second experiment, the error increased to approximately 1.34 m. The trend chart in the figure shows that the AR navigation error linearly increases with the distance traveled. Compared to the first three experimental groups, dynamic lighting and vehicle disturbances were also incorporated during the data collection process in these groups, yet no significant differences in the accuracy attained across the various groups were noted. In the fourth long-distance experiment, significant motion disturbances were introduced alongside variations in lighting conditions and vehicle dynamics. These additional factors contributed to a noticeable degradation in navigation performance. Specifically, the results indicated a marked increase in navigation errors after the introduction of dynamic disturbances, such as abrupt or rapid movements, in the environment. This outcome suggests that while lighting changes and the presence of dynamic objects have a relatively minor impact on the accuracy of AR navigation, intense and unpredictable motion can lead to significant errors in navigation. This is likely due to the difficulty in maintaining stable sensor tracking when subjected to high-speed or erratic movements, which disrupts the continuous localization process. Therefore, during long-distance AR navigation, it is crucial for users to maintain smooth and controlled motion as much as possible, in order to minimize the likelihood of sensor drift and maintain consistent tracking accuracy. This implies that operators should focus on minimizing sudden, severe movements to ensure reliable sensor data, which directly impacts the precision of the navigation system. Furthermore, in the context of extended long-distance navigation, it is essential to periodically incorporate global constraints for relocalization. This process helps mitigate the effects of trajectory drift, which tends to accumulate over long travel distances. By periodically recalibrating the system using these global constraints, the model can correct for any errors in positioning and prevent significant deviation from the true path. This insight underscores a critical area that requires further attention and optimization in future research. Specifically, improving the system's robustness against dynamic disturbances and incorporating more frequent relocalization techniques will be vital for enhancing the overall reliability and accuracy of AR navigation in real-world, long-distance scenarios.

5 | Discussion and Conclusion

This paper proposes an integrated framework for indoor navigation based on end-to-end visual localization and ARCore. The framework introduces an end-to-end visual pose estimation method for global localization in indoor environments. Subsequently, it generates an indoor navigation topology network based on the BIM model and aligns the sparse point cloud obtained from SFM for precise path planning. Finally, real-time indoor navigation is achieved using ARCore. We conducted experiments on the publicly available indoor datasets 7scenes and a self-collected large-scale underground parking dataset to validate the accuracy and robustness of global localization and AR navigation. The experimental results demonstrate that our method achieves sub-20 cm positioning accuracy in the 7scenes dataset, and maintains centimeter-level accuracy in the large-scale parking dataset. Compared to other existing visual localization methods, the method proposed in this paper can achieve higher accuracy in both position and angle. Regarding AR navigation, this can yield stable navigation accuracy without inducing drastic movements. The results show that the navigation error increases with the walking distance and the introduction of severe shaking may further increase the navigation error. Overall, the proposed end-to-end visual localization and navigation framework meets the requirements of indoor AR navigation and achieves sub-10 cm global localization accuracy.

The localization and navigation framework presented in this paper, while effective, has several limitations. The global localization algorithm relies primarily on visual features for position recognition and pose estimation, which can lead to failures in environments with sparse textures. Additionally, due to constraints in data collection viewpoints, significant angular differences between the localization image and the images in the database may result in incorrect scene recognition. These challenges can be addressed by incorporating geometric feature constraints or improving the comprehensiveness of the data collection process. Furthermore, the navigation module predominantly uses a combination of visual data and IMU (Inertial Measurement Unit) for continuous localization, which may lead to trajectory drift over long periods of movement. To mitigate this issue, future work will focus on integrating diverse feature types to create more robust indoor feature maps, thereby improving feature descriptions. We also plan to intermittently apply global localization constraints and local loop closure optimizations during navigation to reduce trajectory drift over extended distances.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

Abaspur Kazerouni, I., L. Fitzgerald, G. Dooly, and D. Toal. 2022. "A Survey of State-of-the-Art on Visual SLAM." *Expert Systems with Applications* 205: 117734.

Acharya, D., K. Khoshelham, and S. Winter. 2019. "BIM-PoseNet: Indoor Camera Localisation Using a 3D Indoor Model and Deep Learning From Synthetic Images." *ISPRS Journal of Photogrammetry and Remote Sensing* 150: 245–258.

Acharya, D., M. Ramezani, K. Khoshelham, and S. Winter. 2019. "BIM-Tracker: A Model-Based Visual Tracking Approach for Indoor Localisation Using a 3D Building Model." 150: 157–171.

14679671, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/tgis.13298 by South University Of Science, Wiley Online Library on [27/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Acharya, D., S. Singha Roy, K. Khoshelham, and S. Winter. 2020. "A Recurrent Deep Network for Estimating the Pose of Real Indoor Images From Synthetic Image Sequences." *Sensors* 20, no. 19: 5492.

Acharya, D., C. J. Tatli, and K. Khoshelham. 2023. "Synthetic-Real Image Domain Adaptation for Indoor Camera Pose Regression Using a 3D Model." *ISPRS Journal of Photogrammetry and Remote Sensing* 202: 405–421.

Acharya, D., R. Tennakoon, S. Muthu, K. Khoshelham, R. Hoseinnezhad, and A. Bab-Hadiashar. 2022. "Single-Image Localisation Using 3D Models: Combining Hierarchical Edge Maps and Semantic Segmentation for Domain." *Adaptation* 136: 104152.

Atia, M. M., A. R. Hilal, C. Stellings, et al. 2017. "A Low-Cost Lane-Determination System Using GNSS/IMU Fusion and HMM-Based Multistage Map Matching." *IEEE Transactions on Intelligent Transportation Systems* 18, no. 11: 3027–3037.

Bagosi, T., and Z. Baruch. 2011. "Indoor Localization by WiFi." In *2011 IEEE 7th International Conference on Intelligent Computer Communication and Processing*, 449–452. Cluj-Napoca, Romania: IEEE.

Besl, P. J., and N. D. McKay. 1992. "A Method for Registration of 3-D Shapes." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 14, no. 2: 239–256.

Bloesch, M., S. Omari, M. Hutter, and R. Siegwart. 2015. "Robust Visual Inertial Odometry Using a Direct EKF-Based Approach." In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 298–304. Hamburg, Germany: IEEE Xpolre.

Campos, C., R. Elvira, R. JJG, J. M. Montiel, and J. D. Tardós. 2021. "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual–Inertial, and Multimap SLAM." *IEEE Transactions on Robotics* 37, no. 6: 1874–1890.

Castle, R., G. Klein, and D. W. Murray. 2008. "Video-Rate Localization in Multiple Maps for Wearable Augmented Reality." In *2008 12th IEEE International Symposium on Wearable Computers*, 15–22. Pittsburg, PA: IEEE.

Chawathe, S. S. 2008. "Beacon Placement for Indoor Localization Using Bluetooth." In *2008 11th International IEEE Conference on Intelligent Transportation Systems*, 980–985. Biejing, China: IEEE.

Chen, C., Y. Chen, H. Q. Lai, Y. Han, and K. R. Liu. 2016. "High Accuracy Indoor Localization: A WiFi-Based Approach." In *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 6245–6249. Shanghai, China: IEEE.

Chen, C., B. Wang, C. X. Lu, N. Trigoni, and A. Markham. 2023. "Deep Learning for Visual Localization and Mapping: A Survey." *IEEE Transactions on Neural Networks and Learning Systems* 35: 17000–17020.

Chen, L., and H. Luo. 2014. "A BIM-Based Construction Quality Management Model and Its Applications." *Automation in Construction* 46: 64–73.

Chen, S., Y. Zhu, and H. Liu. 2023. "PMIndoor: Pose Rectified Network and Multiple Loss Functions for Self-Supervised Monocular Indoor Depth Estimation." *Sensors* 23, no. 21: 8821.

De Blas, A., and I. D. López-de. 2017. "Improving Trilateration for Indoors Localization Using BLE Beacons." In *2017 2nd International Multidisciplinary Conference on Computer and Energy Science (SpliTech)*, 1–6. Split, Croatia: IEEE.

DeTone, D., T. Malisiewicz, and A. Rabinovich. 2018. "Superpoint: Self-Supervised Interest Point Detection and Description." In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 224–236.

Devlin, J., M. W. Chang, K. Lee, and K. Toutanova. 2019. "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding." arXiv. arXiv:1810.04805 [cs]. http://arxiv.org/abs/1810.04805.

Dijkstra, E. W. 1959. "A Note on Two Problems in Connexion With Graphs." *Numerische Mathematik* 1, no. 1: 269–271. https://doi.org/10.1007/BF01386390.

Engel, J., V. Koltun, and D. Cremers. 2017. "Direct Sparse Odometry." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 40, no. 3: 611–625.

Engel, J., T. Schöps, and D. Cremers. 2014. "LSD-SLAM: Large-Scale Direct Monocular SLAM. In: Computer Vision–ECCV 2014." 13th European Conference, Zurich, Switzerland, September 6–12, Proceedings, Part II 13 Springer, 834–849.

Fisher, A., R. Cannizzaro, M. Cochrane, C. Nagahawatte, and J. L. Palmer. 2021. "ColMap: A Memory-Efficient Occupancy Grid Mapping Framework." *Robotics and Autonomous Systems* 142: 103755.

Google. "ARCore Documentation." Accessed March 29, 2023. 2024. https://developers.google.com/ar/develop.

He, K., G. Gkioxari, P. Dollár, and R. Girshick. 2017. "Mask r-cnn." Proceedings of the IEEE International Conference on Computer Vision, 2961–2969.

Heydariaan, M., H. Dabirian, and O. Gnawali. 2020. "Anguloc: Concurrent Angle of Arrival Estimation for Indoor Localization With Uwb Radios." In *2020 16th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, 112–119. Marina del Rey, CA: IEEE.

Huang, B., J. Zhao, and J. Liu. 2020. "A Survey of Simultaneous Localization and Mapping with an Envision in 6G Wireless Networks." arXiv. arXiv:1909.05214 [cs]. http://arxiv.org/abs/1909.05214.

Isikdag, U., S. Zlatanova, and J. Underwood. 2013. "A BIM-Oriented Model for Supporting Indoor Navigation Requirements." *Computers, Environment and Urban Systems* 41: 112–123.

Jang, J. W., and S. N. Hong. 2018. "Indoor Localization With WiFi Fingerprinting Using Convolutional Neural Network." In *2018 Tenth International Conference on Ubiquitous and Future Networks (ICUFN)*, 753–758. Prague, Czech Republic: IEEE.

Jing, L., and Y. Tian. 2020. "Self-Supervised Visual Feature Learning With Deep Neural Networks: A Survey." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, no. 11: 4037–4058.

Kasprzak, S., A. Komninos, and P. Barrie. 2013. "Feature-Based Indoor Navigation Using Augmented Reality." In *2013 9th International Conference on Intelligent Environments*, 100–107. Athens, Greece: IEEE.

Kendall, A., M. Grimes, and R. Cipolla. 2015. "Posenet: A Convolutional Network for Real-Time 6-Dof Camera Relocalization." In *Proceedings of the IEEE International Conference on Computer Vision*, 2938–2946. Santiago, Chile: IEEE Xplore.

Kim, B., W. Bong, and Y. C. Kim. 2011. "Indoor Localization for Wi-Fi Devices by Cross-Monitoring AP and Weighted Triangulation." In *2011 IEEE Consumer Communications and Networking Conference (CCNC)*, 933–936. Las Vegas, NV: IEEE.

Kim, M., J. Koo, and G. Kim. 2023. "Ep2p-Loc: End-to-End 3D Point to 2D Pixel Localization for Large-Scale Visual Localization." In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 21527–21537. Paris, France: IEEE.

Leutenegger, S., P. Furgale, V. Rabaud, M. Chli, K. Konolige, and R. Siegwart. 2013. "Keyframe-Based Visual-Inertial SLAM Using Nonlinear Optimization." *Robotics: Science and Systems IX*. http://www.roboticsproceedings.org/rss09/p37.pdf.

Lou, Y., C. Zhang, Y. Zheng, X. Xie, W. Wang, and Y. Huang. 2009. "Map-Matching for Low-Sampling-Rate GPS Trajectories." In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances*

14679671, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/tgis.13298 by South University Of Science, Wiley Online Library on [27/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in Geographic Information Systems GIS '09, 352–361. New York, NY: Association for Computing Machinery. https://doi.org/10.1145/1653771.1653820.

Luo, R. C., and T. J. Hsiao. 2019. "Indoor Localization System Based on Hybrid Wi-Fi/BLE and Hierarchical Topological Fingerprinting Approach." *IEEE Transactions on Vehicular Technology* 68, no. 11: 10791–10806.

Malekzadeh, P., A. Mohammadi, M. Barbulescu, and K. N. Plataniotis. 2019. "STUPEFY: Set-Valued Box Particle Filtering for Bluetooth Low Energy-Based Indoor Localization." *IEEE Signal Processing Letters* 26, no. 12: 1773–1777.

Manglani, S. 2023. "Real-Time Vision-Based Navigation for a Robot in an Indoor Environment." arXiv preprint. arXiv:230700666.

Middelberg, S., T. Sattler, O. Untzelmann, and L. Kobbelt. 2014. "Scalable 6-Dof Localization on Mobile Devices." In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, Proceedings, Part II 13 Springer*, 268–283. Zurich, Switzeerland: Springer International Publishing.

Montemerlo, M., S. Thrun, D. Koller, and B. Wegbreit. 2002. "FastSLAM: A Factored Solution to the Simultaneous Localization and Mapping Problem." AAAI-02 Proceedings.

Mur-Artal, R., J. M. M. Montiel, and J. D. Tardos. 2015. "ORB-SLAM: A Versatile and Accurate Monocular SLAM System." *IEEE Transactions on Robotics* 31, no. 5: 1147–1163.

Newson, P., and J. Krumm. 2009. "Hidden Markov Map Matching Through Noise and Sparseness." In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems GIS '09*, 336–343. New York, NY: Association for Computing Machinery. https://doi.org/10.1145/1653771.1653818.

Prorok, A., and A. Martinoli. 2014. "Accurate Indoor Localization With Ultra-Wideband Using Spatial Models and Collaboration." *International Journal of Robotics Research* 33, no. 4: 547–568.

Ranganathan, A., S. Matsumoto, and D. Ilstrup. 2013. "Towards Illumination Invariance for Visual Localization." In *2013 IEEE International Conference on Robotics and Automation*, 3791–3798. Kalsruhe, Germany: IEEE.

Sarlin, P. E., A. Unagar, M. Larsson, et al. 2021. "Back to the Feature: Learning Robust Camera Localization From Pixels to Pose." In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 3247–3527.

Sattler, T., W. Maddern, C. Toft, et al. 2018. "Benchmarking 6DOF Outdoor Visual Localization in Changing Conditions."

Smith, R., M. Self, and P. Cheeseman. 1987. "A Stochastic Map for Uncertain Spatial Relationships." *Machine Intelligence and Pattern Recognition*: 467–474. Cambridge, MA: MIT Press.

Song, X., X. Fan, X. He, et al. 2019. "Cnnloc: Deep-Learning Based Indoor Localization With Wifi Fingerprinting." In *2019 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*, 589–595. Leicester, UK: IEEE.

Tang, S., Q. Zhu, W. Wang, and Y. Zhang. 2015. "Automatic Topology Derivation From IFC Building Model for In-Door Intelligent Navigation." *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* 40, no. 4: 7–11.

Teo, T. A., and K. H. Cho. 2016. "BIM-Oriented Indoor Network Model for Indoor and Outdoor Combined Route Planning." *Advanced Engineering Informatics* 30, no. 3: 268–282.

Ummenhofer, B., H. Zhou, J. Uhrig, et al. 2017. "Demon: Depth and Motion Network for Learning Monocular Stereo." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 5038–5047. Honolulu, HI: IEEE Xplore.

Vaswani, A., N. Shazeer, N. Parmar, et al. 2017. "Attention Is All You Need." arXiv. arXiv:1706.03762 [cs]. http://arxiv.org/abs/1706.03762.

Veličković, P., G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. 2018. "Graph Attention Networks." arXiv. arXiv:1710.10903 [cs, stat]. http://arxiv.org/abs/1710.10903.

Wang, Q., J. Guo, and M. K. Kim. 2019. "An Application Oriented Scan-to-BIM Framework." *Remote Sensing* 11, no. 3: 365.

Ying, C., T. Cai, S. Luo, et al. 2021. "Do Transformers Really Perform Bad for Graph Representation?" arXiv. arXiv:2106.05234 [cs]. http://arxiv.org/abs/2106.05234.

Zafari, F., A. Gkelias, and K. K. Leung. 2019. "A Survey of Indoor Localization Systems and Technologies." *IEEE Communications Surveys & Tutorials* 21, no. 3: 2568–2599.

Zhang, T. Y., and C. Y. Suen. 1984. "A Fast Parallel Algorithm for Thinning Digital Patterns." *Communications of the ACM* 27, no. 3: 236–239.

Zhang, W., G. Liu, and G. Tian. 2019. "A Coarse to Fine Indoor Visual Localization Method Using Environmental Semantic Information." *IEEE Access* 7: 21963–21970.

Zhou, X., Q. Xie, M. Guo, J. Zhao, and J. Wang. 2020. "Accurate and Efficient Indoor Pathfinding Based on Building Information Modeling Data." *IEEE Transactions on Industrial Informatics* 16, no. 12: 7459–7468.

Zhuang, Y., J. Yang, Y. Li, L. Qi, and N. El-Sheimy. 2016. "Smartphone-Based Indoor Localization With Bluetooth Low Energy Beacons." *Sensors* 16, no. 5: 596.

14679671, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/tgis.13298 by South University Of Science, Wiley Online Library on [27/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License